

Properties of Schemes and Morphisms

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Contents

1	Properties of Schemes	1
2	Properties of Morphisms	3
3	Diagonals	5

1 Properties of Schemes

Definition 1.1. A scheme X is **quasicompact** (qc) when any open cover has a finite subcover.

X is **quasiseparated** (qs) if the intersection of any two quasicompact open subsets is quasicompact.

Proposition 1.2. Let X be a quasicompact scheme. Then, any $P \in X$ specializes to a closed point $\mathfrak{m}$, i.e., there exists a closed point $\mathfrak{m} \in \overline{\{P\}}$.

Proof. Let P be a point of X . We want to show the existence of a “maximal” specialization of P , so we consider the poset of specializations. That is, consider the set

$$\overline{\{P\}}$$

which is all points that are specializations of P , and define an order via

$$a \preceq b \iff b \in \overline{\{a\}}$$

(which is more usually written if $a \rightsquigarrow b$). To find a maximal element with respect to this order, we apply Zorn’s lemma.

To apply, first we see that $\overline{\{P\}} \neq \emptyset$, since it contains P . Second, let $\mathcal{A}$ be a chain in $\overline{\{P\}}$ (a totally ordered subset). To argue that this is bounded above, we observe that

$$\bigcap_{a \in \mathcal{A}} \overline{\{a\}} \neq \emptyset,$$

since an intersection of nested nonempty closed sets is nonempty in a quasicompact space (the usual topology fact), so we can choose a point b in the intersection which will be a mutual specialization of all such $a \in \mathcal{A}$.

Therefore, Zorn's lemma implies there is a maximal specialization $\mathfrak{m}$, which is exactly a closed point specializing P . $\square$

Definition 1.3. A property P of schemes is called **local** when

- (a) if X has P , then all open subsets U have P
- (b) if $\bigcup U_i = X$ is an open cover where each U_i has P , then X has P .

Definition 1.4. A property P of schemes is called **affine local** when there is a property Q of affine opens where X has P when there is a cover of affine opens having Q , and furthermore Q satisfies

- (a) if $\text{Spec } A \hookrightarrow X$ has Q , then $\text{Spec } A_f \hookrightarrow X$ has Q .
- (b) if $(f_1, \dots, f_n) = A$ (i.e., the $\text{Spec } A_{f_i}$ cover $\text{Spec } A$) and $\text{Spec } A_{f_i} \hookrightarrow X$ has Q for all i , then so does $\text{Spec } A \hookrightarrow X$.

In the way that a local property P being true for a scheme X implies that P is true for any open U , we would ask something analogous for an affine local property P .

Lemma 1.5 (Affine Communication Lemma). Let P be an affine local property, where X has P if there exists an affine open cover $\bigcup \text{Spec } A_i$ where each $\text{Spec } A_i$ has Q . In this case, if X has P , then every affine open (not just those in a cover) has Q .

In particular, the Affine Communication Lemma implies that affine local properties are local, if X has P , then any open $U \subseteq X$ has P because any affine open cover of U consists of affine open subsets of P , which will then have Q .

Example 1.6. The following properties are affine-local

- Any **stalk-local property** (a scheme has property P iff all stalks have property Q)
 - **reduced** (all stalks reduced)
 - **normal** (all stalks integrally closed)
 - **factorial** (all stalks UFDs)
- **Locally Noetherian**, that X every affine open set $\text{Spec } A$ has A is Noetherian.
- **Locally finite type A -scheme**, that $X \rightarrow \text{Spec } A$ is an A -scheme, with every affine open $\text{Spec } B \rightarrow \text{Spec } A$ has B a finitely generated A -algebra.
- An $\mathcal{O}_X$ -module $\mathcal{F}$ being **quasicoherent**, that restricted to any affine open $\text{Spec } A$, $\mathcal{F}|_{\text{Spec } A} \cong \widetilde{M}$ for an A -module.
- A quasicoherent sheaf $\mathcal{F}$ being **finite type**, **finitely presented**, or **coherent**, which is that for every affine open $\text{Spec } A$, $\Gamma(\text{Spec } A, \mathcal{F})$ is finitely generated, finitely presented,

or coherent respectively. (Recall, an A -module M is coherent if for any morphism $N \rightarrow M$, the kernel is finitely generated).

Definition 1.7. When we name a property “locally P ”, then we will define the property P to be the schemes which are locally P and also quasicompact.

2 Properties of Morphisms

Definition 2.1. If P is a property of schemes, we (typically) say a morphism $f : X \rightarrow Y$ has P when for every affine open subset $U \subseteq Y$, $\pi^{-1}(U)$ has P .

In particular this is how we will define what it means for a morphism to be P = quasicompact, quasiseparated, affine, etc.

Definition 2.2. A class P of morphisms is called **reasonable** if it contains all isomorphisms and the following hold.

- (i) The class is **preserved by composition**: if $\pi : X \rightarrow Y$ and $\rho : Y \rightarrow Z$ are both in this class, then so is $\rho \circ \pi$.
- (i) The class is **preserved by “base change”**: if $\pi : X \rightarrow Y$ is in the class, then for any $Y' \rightarrow Y$, the induced map $X \times_Y Y' \rightarrow Y'$ is in the class.
- (i) The class is **local on the target**:
 - (a) if $\pi : X \rightarrow Y$ is in the class, then for any $V \subseteq Y$ open, $\pi^{-1}(V) \rightarrow V$ is in the class.
 - (a) for $\pi : X \rightarrow Y$, if there is an open cover $\{V_i\}$ of Y for which each restricted morphism $\pi^{-1}(V_i) \rightarrow V_i$ is in the class, then π is in the class.

Definition 2.3. We say a property P of morphisms is **local on the source** when the following holds.

- (a) if $X \rightarrow Y$ has P , then for any open subset, $U \rightarrow X \rightarrow Y$ has P .
- (b) if there is an open cover $\{U_i\}$ of X for which each restricted morphism $U_i \hookrightarrow X \rightarrow Y$ has P , then π has P .

Definition 2.4. We will say P is **affine-local on the target** when there is a property Q of morphisms to affine schemes, where $\pi : X \rightarrow Y$ has P when there is an open cover $\{V_i\}$ of Y where each $\pi^{-1}(V_i) \rightarrow V_i$ has Q , and furthermore Q satisfies properties analogous to (a) and (b) for affine-local properties, i.e.,

- (a) if $\pi^{-1}(\text{Spec } A) \rightarrow \text{Spec } A$ has Q , then $\pi^{-1}(\text{Spec } A_f) \rightarrow \text{Spec } A_f$ has Q .
- (b) if $(f_1, \dots, f_n) = A$ (i.e., the $\text{Spec } A_{f_i}$ cover $\text{Spec } A$) and each $\pi^{-1}(\text{Spec } A_{f_i}) \rightarrow \text{Spec } A_{f_i}$ has Q , then so does $\pi^{-1}(\text{Spec } A) \rightarrow \text{Spec } A$.

By the Affine Communication Lemma, if a property is affine-local on the target, then it is in particular local on the target.

Analogously, one can define what it means for a class of morphisms to be **affine-local on the source**.

Proposition 2.5. Let P be a property preserved by composition and base change, and say $X \rightarrow Y$ and $X' \rightarrow Y'$ are morphisms of S -schemes with property P . Then, $X \times_S X' \rightarrow Y \times_S Y'$ has property P .

Proof. We build the map $X \times_S X' \rightarrow Y \times_S Y'$ in steps. Consider the diagram

$$\begin{array}{ccccccc}
 & & X \times_S X' & \longrightarrow & X \times_S Y' & \longrightarrow & Y \times_S Y' \\
 & \swarrow & & & \swarrow & & \swarrow \\
 X' & \longrightarrow & Y' & & X & \longrightarrow & Y
 \end{array}$$

and we claim that the left/right parallelograms are Cartesian. We just show the left parallelogram is Cartesian. This amounts to showing that there is a bijection of commuting squares

$$\begin{array}{ccc}
 T & \dashrightarrow & X \times_S Y' \\
 \downarrow & & \downarrow \\
 X' & \longrightarrow & Y'
 \end{array}
 \iff
 \begin{array}{ccc}
 T & \dashrightarrow & X \\
 \downarrow & & \downarrow \\
 X' & \longrightarrow & Y'
 \end{array}$$

where starting with $T \rightarrow X \times_S Y'$, we get $T \rightarrow X$ by post composing with the projection, and conversely starting with $T \rightarrow X$, we get $T \rightarrow X \times_S Y'$ by assembling $T \rightarrow X$ and $T \rightarrow X' \rightarrow Y'$. This will imply that our parallelogram really is Cartesian.

Now, since $X' \rightarrow Y'$ has P by assumption, the map $X \times_S X' \rightarrow X \times_S Y'$ has P because it is the base change along a morphism $X \times_S Y' \rightarrow Y'$.

Similarly, we will argue that $X \times_S Y' \rightarrow Y \times_S Y'$ has P , so then the composition has P . $\square$

Definition 2.6. A morphism $\pi : X \rightarrow Y$ is an **open embedding** if π factors as

$$(X, \mathcal{O}_X) \xrightarrow{\cong} (U, \mathcal{O}_Y|_U) \hookrightarrow (Y, \mathcal{O}_Y).$$

Definition 2.7. A morphism $\pi : X \rightarrow Y$ is a **closed embedding** if it is an affine morphism, and for every affine open $\text{Spec } B \subseteq Y$ with $\pi^{-1}(\text{Spec } B) = \text{Spec } A$, the map $B \rightarrow A$ is surjective (so modeled on $B \rightarrow B/I$).

Proposition 2.8. Open embeddings and closed embeddings are reasonable classes of morphisms.

3 Diagonals

Definition 3.1. Let P be a class of morphisms. We define the class of morphisms $P\delta$ (read P -diagonal), where we say $\pi : X \rightarrow Y$ lies in $P\delta$ if $\delta_\pi : X \rightarrow X \times_Y X$ lies in P .

Theorem 3.2 (Cancellation Theorem). Let P be a class of morphisms preserved by base change and composition (e.g. a reasonable class). Suppose

$$\begin{array}{ccc} X & \xrightarrow{\pi} & Y \\ & \searrow \tau & \swarrow \rho \\ & Z & \end{array}$$

is a commuting diagram of schemes. Suppose ρ is in $P\delta$ and τ is in P . Then, π is in P .

We will write for short,

$$\begin{array}{ccc} X & \xrightarrow{\therefore \in P} & Y \\ & \searrow \in P & \swarrow \in P\delta \\ & Z & \end{array}$$

Proof.

□